

Integrated Routing and Grooming in GMPLS-Based Optical Networks

Ding Zhemin and Mounir Hamdi
Department of Computer Science
Hong Kong University of Science and Technology
Clear Water Bay, Kowloon, Hong Kong
Email: hamdi@cs.ust.hk

Abstract

This paper proposes a novel integrated routing and grooming algorithm for IP over WDM networks. Given a peer model in GMPLS-Based optical networks, we take into account the combined topology and resource usage information on both the IP and WDM layers. Based on a clustering technique called the *Blocking Island Paradigm*, we propose an enhanced Blocking Island Graph (BIG) network model with a Blocking Island Hierarchy (BIH) to abstract network resources. The main idea of the algorithm is to keep the integrity as well as balance the load of related Blocking Islands. We also devise a cost function in the routing algorithm to groom the traffic flows into active lightpaths. The complexity of the algorithm is analyzed to show its practicality. In the simulation, we compare our proposed algorithm with three other integrated routing algorithms in terms of blocking probability. The three algorithms are: the integrated min-hop (IMH) routing algorithm, the maximum open capacity (MOCA) routing algorithm and the IP-WDM grooming (IWG) algorithm. The results demonstrate that our algorithm achieves the best performance.

1. Introduction

With the continuous growth of IP traffic, it is widely believed that the coupling of the IP layer and the optical layer will be the major way of transporting long distance traffic in the next generation Internet. This “close coupling” concept evolves from today’s four-layer core network architecture, which transports IP traffic over ATM (Asynchronous Transfer Mode), then over SONET/SDH, and finally over Optical fibers. In this approach, the ATM layer is used because of its capability for traffic engineering and flexible network provisioning. The SONET/SDH layer is used for its transportation capability while optical layer provides the raw high bandwidth of optical fibers. At first thought, this approach makes sense. But the architecture suffers from significant inter-layer overhead and function overlap, which limit the scalability of the whole network. With the appearance of high performance routers and intelligent optical networks, a simpler and more efficient network design is being sought. In this paper, we focus on a network employing IP directly over optical networks, which is the most efficient solution among all the possible approaches.

In the new architecture, the optical layer is usually referred to as optical networks using wavelength-division-multiplexing (WDM) technology. WDM is an optical multiplexing technique which divides a single fiber into multiple channels (wavelengths). Each channel then can be independently modulated to accept different traffic streams. WDM networks have been widely deployed in the backbone, metro, regional and access networks. The function of WDM networks has already gone beyond the simple provision of point-to-point physical links. More intelligent network management is made possible by the introduction of new equipment such as optical cross connects (OXC) and optical burst switchers (OBS).

The integration of IP over WDM involves many issues. One of them is the need to fast and efficiently deploy bandwidth guaranteed end-to-end paths between ingress-egress routers. Motivated by this need, we propose an integrated routing and grooming algorithm for IP over optical networks which takes into account the combined knowledge of the two layers and aggregate different traffic flows to efficiently use the network resources. This integration scheme is more robust and is more flexible to changing traffic than previous schemes which only employ dynamic routing in the IP layer and use a static wavelength routed network

topology underneath. Notice the two-layer routing without good coordination is prone to several problems. One of them is that the bandwidth of some wavelengths may be occupied by a small amount of traffic and is very much underused. This situation could be prevented if proper traffic grooming is carried out on the upper layer. On the other side, the congested routers are the main reason for the reduced performance in the IP layer. The congestion of routers in the IP layer can be reduced by adjusting the virtual topology of WDM networks.

In the proposed algorithm, we assume the bandwidth requirement can be a fraction of a lightpath and the interconnection topology of the two layers can change dynamically. We also assume that similar control planes are used in the IP and WDM networks. Specially, the recent advances in *generalized multiprotocol label switching* (GMPLS) [6], proposed by the Internet Engineering Task Force (IETF), have provided an enhanced signaling and routing mechanism in both the IP and WDM layers, which makes it possible for the implementation of our integrated routing and grooming algorithm.

The remainder of the paper is organized as follows: we firstly introduce different IP over WDM models and present a brief survey on various integrated routing algorithms. In section 3, we define the routing problem and network model based on GMPLS. A paradigm called Blocking Island and an integrated routing and grooming algorithm based on this paradigm are discussed in section 4 and 5. In section 6, we compare and analyze the simulation results of different routing algorithms. Finally a conclusion of this paper is given.

2. Related Work

From an architectural point of view, optical networks provide data transport channels with high bandwidth while IP routers at the edge of optical networks are responsible to set up the end-to-end routing path before any communication takes place. Although it is agreed that an IP control plane over optical networks is necessary to provide certain functionalities such as signaling and routing, there are different views on how information is shared between the two layers and how IP routers interact with optical nodes. A general IP over WDM networking model is shown in Figure 1, where multiple optical sub-networks are interconnected by well defined signaling and routing interface, also known as the network-network interface (NNI), to function as a core optical network. Clients (IP/GMPLS routers) are attached to the optical core network, setting up end-to-end paths over the virtual topology defined using optical lightpaths. The signaling and routing information between the clients and the core optical networks is interchanged by the user-network interface (UNI).

Based on different degrees of information sharing and control sharing between the two layers, we have three interconnection models [1]: overlay model, augmented model and peer model.

Overlay model: In the overlay model, each layer is independent and retains its own protocols. The communication between the two layers is minimal and is handled in a “client-server” way. The advantage of this model is that all the solutions (such as traffic engineering, routing and signaling protocols and so on) developed in each layer can be applied directly without any modification. It is the most practical model in the near future for deployment. The disadvantage of this model is that there is no information or control sharing between the two layers. The optical layer has to be statically configured and the connectivity has to be fixed before any route is set up in the IP layer. Fast provisioning and restoration is impossible. The two step-routing also fails to dynamically adjust to the constantly altering resource availability.

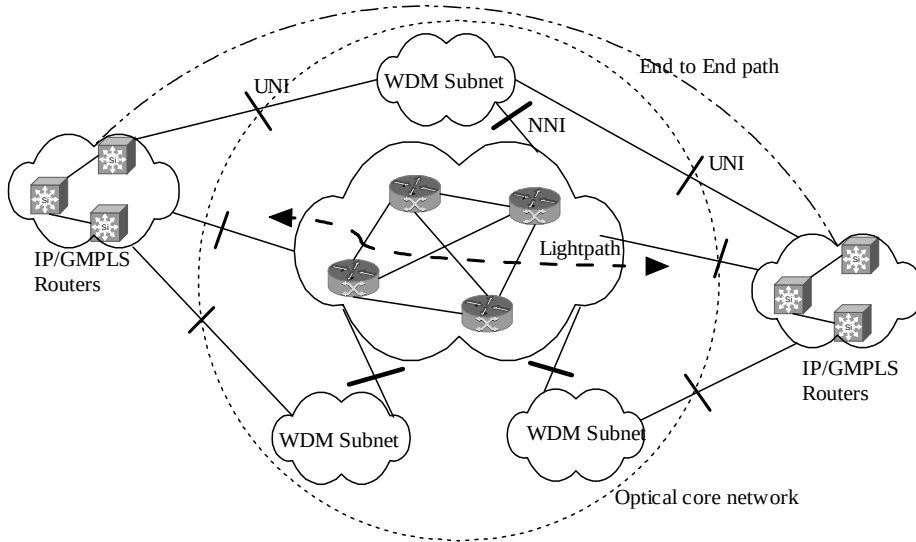


Figure1: A general IP over WDM networking model

Augmented model: Aware of the inefficiency and inflexibility because of total independence of the two layers, the augmented model allows certain information sharing between the IP and the optical layers while each layer still keeps separated routing protocols. For example, external IP addresses could be carried within the optical routing protocols to allow reachability information to be passed to IP clients.

Peer model: By deploying a single integrated control plane over the optical layer and the IP layer, we have the peer model. Under this model, two layers are treated together and a set of uniformed protocols is used. The resource allocation and management is made in a unified way. Unlike the previous two models where resource utilization is optimized independently in each layer, the goal of global optimization can be achieved in a peer model. The advantage of this model is the information transparency and seamless interconnection of the IP and the optical layers. The disadvantage is the shortage of equipment and standards. This tight integration may not be available in the near future.

The routing approaches for three models are different. The integrated routing strategy is best supported by a peer model, whose one integrated control plane is more flexible and efficient and more capable to provide the fast provision/restoration service. Although the overlay model and augmented model is more feasible for the near future deployment, in the long term, the peer model is a better solution with technology advancement in the relevant equipment.

In this paper, we study the integrated routing and grooming algorithm under the peer model assumption. We firstly present a brief survey on the available integrated routing algorithms. A framework is proposed in [1] to use Open Shortest Path First (OSPF) [7] with “suitable” optical extensions. The assumption is that link state information is maintained by all nodes and IP routers and OXCs are treated equally in a joint network topology. In this framework, although the optical layer seems transparent to the upper IP layer, the IP routing and lightpath routing are still carried out separately as a two-step process, failing to make full use of the combined information. The algorithm proposed in [2] focuses on grooming. By incorporating the information of incoming traffic flows with load distribution of lightpaths, a constraint-based routing algorithm is developed. The main objective of this algorithm is to optimize the resource usage by grooming a new flow into an already existing lightpath. A cost function is designed to measure the possible grooming. It is difficult to find the appropriate cost coefficients, which are derived in [2] from experimental results. [3] proposes an algorithm for integrated dynamic routing of bandwidth guaranteed paths in IP over WDM

networks. The key idea is to pick paths that don't interfere too much with potential future traffic flows. The interference is estimated by computing the maximum flow between ingress-egress routers. The algorithm takes into account the combined knowledge of resource and topology information in both the IP and the optical layers. However, the computation of the algorithm is quite expensive and it assumes the information of ingress/egress nodes is already known.

3. Problem definition and network model based on GMPLS

In this section, a possible IP over WDM architecture based on GMPLS is presented. GMPLS is a generalized MPLS architecture to include non-packet-based control planes, as well as the conventional packet networks. In GMPLS, proposals have been made to extend the base function of RSVP (Resource Reservation Protocol) and LDP (Label Distribution Protocol) to enhance the signaling [8]. GMPLS also supports link bundling by which an IP link can consist of a number of WDM fibers. It is possible to aggregate multiple TE (Traffic Engineering) labels in a bigger TE-LSP (Label Switched Path) to achieve grooming benefits [6]. LMPs (Link Management Protocols) are included in GMPLS for neighborhood discovery in optical networks. Under the assumption of peer model, the link-state and resource usage information are distributed to both layers. Notice in our algorithm, the only dynamic information needed is the bandwidth availability in the IP layers and wavelength availability in the WDM layers.

We define a network topology $G(V, L, W)$ for a given IP/WDM network, where V is the set of all nodes; L is the set of bidirectional optical links and W is the set of wavelengths per fiber link. Here we assume the number of wavelengths on each fiber link to be the same. Under the peer model assumption, network nodes are treated as integrated router/OXC nodes and there is only one control plane. While in practice, it is possible some nodes, which are only OXCs without the function of IP routers, to remain part of an IP/WDM network. Therefore we consider $V(R, O)$ for a given set of nodes in an IP/WDM optical network, where R is the set of integrated router/OXC nodes, O is the set of OXC nodes. Nodes in R can multiplex or demultiplex traffic requests with any bandwidth granularity and do wavelength conversion. Nodes in O can only multiplex or demultiplex traffic requests with the bandwidth granularity of a whole wavelength and don't have wavelength conversion capability. The transaction power of each node is only limited by the network resource availability. This assumption can be relaxed if the detailed transmission capability of equipment is given. An example of network topology is shown in Figure 2, where node 1, 2, 3, 4, 8 are integrated router/OXC nodes and nodes 5, 6, 7 are plain OXCs.

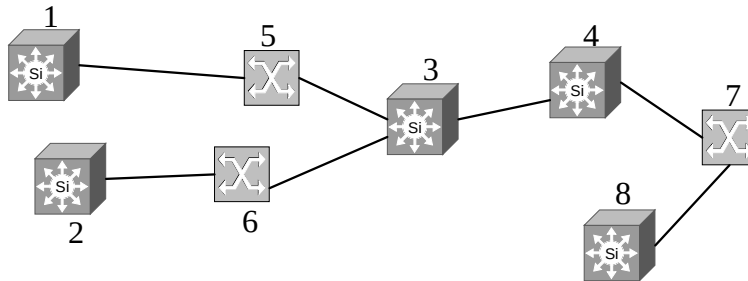


Figure 2: An example of IP over WDM network

Based on the GMPLS framework, an optical channel (λ -LSP) needs to be set up for each request and the required bandwidth is reserved on links of the λ -LSP path. The request to set up a λ -LSP can be defined as $(X_\mu, Y_\mu, \beta_\mu)$ where X_μ and Y_μ are distinct nodes of the network; β_μ is the required bandwidth. Since this is a circuit switched network, the only QoS requirement we consider in this paper is bandwidth. The main

objective of our algorithm is to find a bandwidth guaranteed path, either in the existing IP layer or in a new topology by setting up new wavelength paths. Assume the bandwidth of a whole wavelength is 1 unit. A request (X_u, Y_u, β_u) is to be routed from node $X_u \in R$ to node $Y_u \in R$ with the bandwidth requirement $\beta_u \leq 1$ unit. If an optical channel is set up to reach the destination and this path involves nodes of OXCs, some cut through arcs (lightpaths) may be created to meet the requirement. The IP layer network topology will be changed in this case. For example, in figure 2, a traffic request arrives, requiring a bandwidth of 0.5 unit from node 1 to node 8. To simplify the example, we consider each fiber to have only one wavelength. Assume an LSP path $(1 \rightarrow 5 \rightarrow 3 \rightarrow 4 \rightarrow 7 \rightarrow 8)$ has been found from node 1 to node 8 along the wavelength w_1 . Because OXCs can only multiplex and demultiplex traffic requests with a bandwidth request of a whole wavelength, new lightpaths are set up to directly connect integrated nodes. In figure 3, 2 new lightpaths (cut through arcs) are introduced to form a new topology. Notice that only 0.5 unit bandwidth is consumed along the path. The residual 0.5 unit bandwidth is still available along the lightpath for future use. Those lightpaths are logical links in the IP layer. They can be released or re-setup according to the traffic requests and the resource availability.

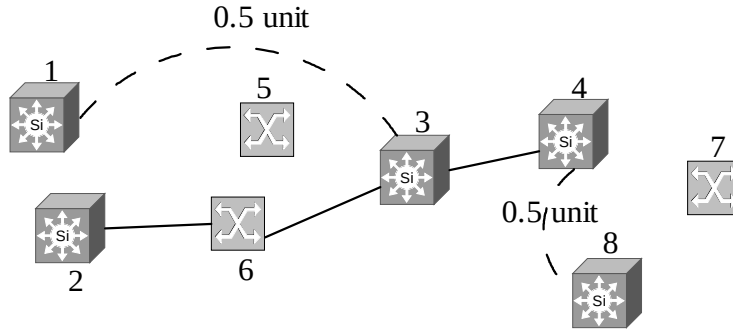


Figure 3: A new topology of the example network.

4. Blocking Island Paradigm and *MS* heuristic

In this section, we give an introduction on the paradigm used in our routing and grooming algorithm. The Blocking Island paradigm proposed in [4] provides an efficient way of abstracting resource (especially bandwidth) available in a communication network.

The blocking Island clusters parts of network according to the bandwidth availability. A β -BI (β -Blocking Island) for a node x is the set of all nodes of the network that can be reached from x using links with at least β available bandwidth. For example, Figure 4 (a) is a network topology of NSFNet where the available bandwidth of each link is given in brackets. $L_n(x)$ refers to a link n which has x available bandwidth. $N_1 = (V_1, V_2, V_3, V_4)$ is a 40-BI for node V_1 , which means any other node is not able to find a path with at least 40 available bandwidth to reach V_1 . β -BI has some very useful properties. Below we list a few without proof (for a proof, please refer to [4]).

Unicity: there is one and only one β -BI for a node. Thus if S is the β -BI for a node, S is the β -BI for every node in S .

Partition: β -BI induces a partition of nodes in a network.

Route existence: give a request $d_u = (x_u, y_u, \beta_u)$, it can be satisfied if and only if the node x_u and y_u are in the same β_u -BI.

Inclusion: If $\beta_i < \beta_j$, the β_j -BI for a node is a subset of the β_i -BI for the same node.

By using the BI concept, we may identify the bottle links easily. For example, after constructing a β -BI, we can safely conclude that the links of this β -BI's co-cycle do not have enough available resources to support a demand requiring β (or more) bandwidth. Given a traffic request with β (or more) bandwidth requirement, if the request can be satisfied, then the source node and the destination node must be in the same β -BI and the links of the route (if it exists) must be in the β -BI. However, it is not true that all links inside the β -BI have at least β available bandwidth. Finally, if the bandwidth requirement is 0, it will be a Blocking Island including the whole network.

We use these properties to build a β -BI graph, which is a simple graph representing an abstraction view of the available network resources: each β -BI is clustered into a single node and there are abstract links between two of these nodes. These abstract links denote the critical links, since their available bandwidth is not enough to support a demand requiring β resources. For example, figure 4 (b) is a 40-BI graph of the network topology shown in Figure 4 (a). The bandwidth of those abstract links between clustering nodes is less than 40.

Using the concept β -BIG, a recursive decomposition of Blocking Island Graphs in decreasing order of β s, e.g. $\beta_1 > \beta_2 > \dots > \beta_n$ can be constructed. We call this layered structure of Blocking Island Graphs (BIGs) a Blocking Island Hierarchy (BIH). For example, according to the demand $40 > 20 > 10$, we have such a BIH: Figure 4 (b) is a 40-BIG, Figure 4(c) is a 20-BIG and figure 4(d) is a 10-BIG.

With this abstract technique, instead of investigating the whole network topology, we focus our attention only on a small part. For example, given a traffic request $(V_1, V_4, 40)$ in figure 4, according to the BIH, the route is in 40-BI N_1 if it exists. In the N_1 Blocking Island, different routing heuristics can be employed to find the route. If the bandwidth requirement is not the exact match of the available blocking island graphs, one possible solution is to build a β -BIG (β is the required bandwidth) but it is time-consuming and in some cases, it may not be feasible. So we instead look at the blocking island graph with the closest value. For example, given a traffic request $(V_1, V_4, 50)$, based on BI properties, we know the route must be in 40-BI N_1 if it is available. The search space is probably a little bigger than that of a 50-BIG while it is still much smaller than the whole network topology.

If the route is allocated, the available link capacity is decreased and the BIH may need to be modified. For example, in figure 4(a), if we assign a route $V1 \rightarrow V3 \rightarrow V2$ with 40 bandwidth, the 40-BI N_1 will be split into two 40-BIs: $(V1, V2, V4)$ and $(V3)$. Notice, all the modifications are actually localized and carried out only within the N_1 Blocking Island. After the splitting, some requests that can be satisfied before the allocation of the route cannot be satisfied anymore. Based on analysis of the consequences of the splitting, a heuristic called *Minimum Splitting (MS)* is proposed. The difference between this heuristic and the others is that it tries to find a route which doesn't provoke a split in the corresponding BI graph. If the splitting is unavoidable, we would rather select a route which incurs the fewest splitting in the BI graph. We also look at the BIH to make sure the splitting will cause minimum impact on other layers in the hierarchy, especially layers lower (with a smaller β) than the current BIG. The rationale behind the heuristic is simple: the more splitting, the more links become critical, the more possible a future traffic request will be blocked. By always picking up the route causing minimum splitting, we try to reserve the largest resource possible for the future request and balance the load among all nodes.

The heuristic can be understood by a simple example. An IP over WDM network is shown in figure 5, where every ingress or egress node is composed of an IP router and an OXC in a peer configuration. Node 5 is equipped only with an OXC and cannot do any sub-wavelength granularity switching or multiplexing. We assume each fiber link carries two wavelengths, λ_1 and λ_2 . By using the BIG model proposed in [5], the

network topology is transformed to figure 5 (b), where each wavelength has an identical duplication of the original topology. Because the integrated router/OXC nodes have the capability of wavelength conversion, we modify the BIG model by adding virtual links to represent the increasing connectivity between each node pair. The weight (distance) of those virtual links (shown in figure 5 in dash lines) is 0.

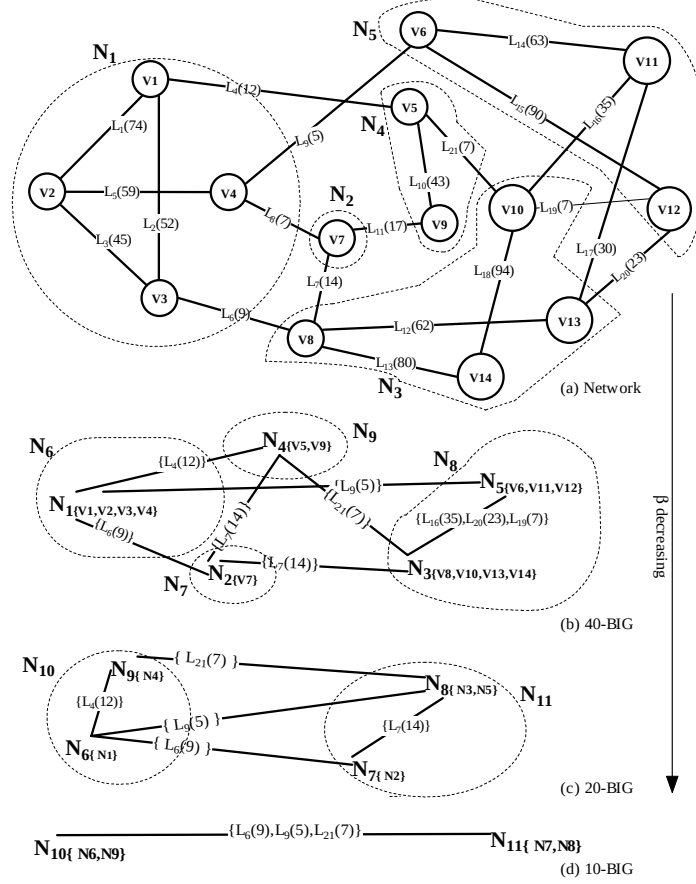


Figure 4: The blocking island hierarchy for bandwidth requirement $\{40, 20, 10\}$. (a) Network graph; (b) 40-BIG; (c) 20-BIG; (d) 10-BIG

Given a request for 0.2 wavelengths from node 4 to node 3, since there is no traffic at any wavelength topology we simply route it along the wavelength λ_1 . Note, at the end of the routing, there will be 0.8 units of residual bandwidth on the cut through arc between nodes 4 and 3. Figure 6 (a) shows the combined state of the IP layer and the optical layer after the previous request is routed. A new lightpath has been set up as a cut through arc between node 4 and node 3 with a residual bandwidth of 0.8 units in the IP layer. As a result, some fiber links have been deleted in the optical layer. Assume another traffic request arrives needing 0.2 wavelengths from node 1 to node 2. If only the shortest path is used, either route in λ_1 or λ_2 will be the same. However, if we use *Minimum Splitting* heuristic here (the hierarchy is only 1 level in this example), we find the route in λ_2 provokes a splitting and therefore disconnect node 1 and node 2 from node 3 and node 4. The resulted splitting is shown in figure 6 (b). On the other hand, a route in λ_1 will keep the integrity and connectivity of the whole topology. So it is better to choose the route in λ_1 .

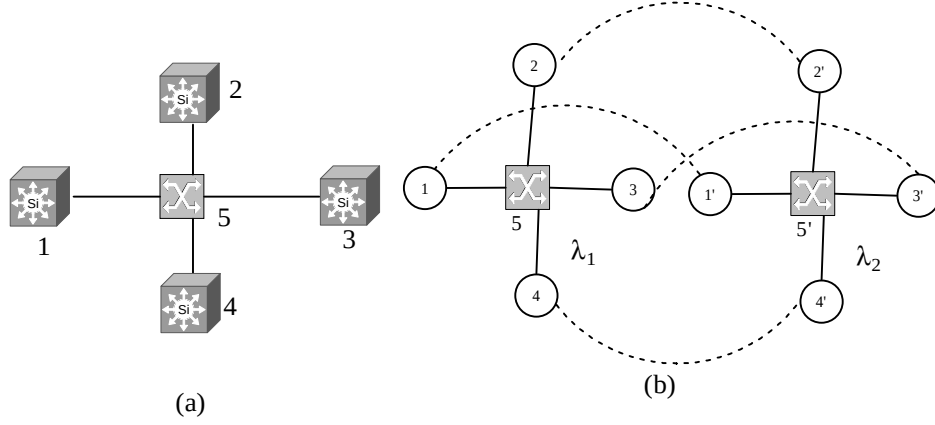


Figure 5: (a) Illustrative Example network (b) Representation of the network by BIG model

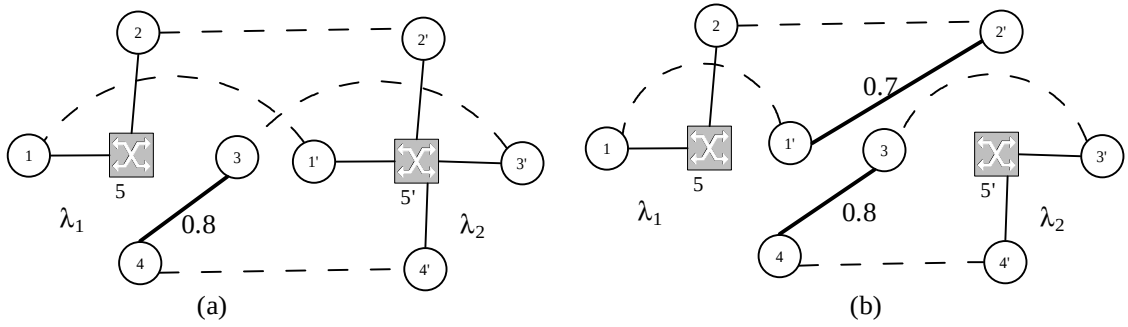


Figure 6: (a) Graph after routing $4 \rightarrow 3$; (b) the splitting caused by a route from node 1 to node 2 in λ_2

5. Integrated Routing and Grooming Algorithm Based on Blocking Island Paradigm

In this section, before presenting a formal description of the algorithm, we outline the key ideas used in our method. As illustrated in the example (figure 5), in order to apply the BI paradigm into our algorithm, the first step is to transform the original network topology into a set of blocking island graphs. In [5], we propose a BIG network model to represent WDM optical networks. It is not appropriate to apply this model directly since there are some differences between the modeling of IP over WDM networks and WDM networks. In the original BIG model, we assume a single fiber network without wavelength converters. Each connection request needs to be allocated over a route and assigned one wavelength. It is modeled as a simplified blocking island graph with only one level of BIH. For IP over WDM networks, the integrated router/OXC nodes have the capability of wavelength conversion. The traffic requests can require any fraction of wavelength bandwidth. And it will have multi-level layers of BIH according to different traffic requests.

An Enhanced BIG Network Model

To accommodate those variations, we propose an enhanced BIG network model to represent the IP over WDM network. We assume a single fiber optical network and the number of wavelengths on each fiber link to be the same. Consider a network topology $G(V, L, W)$ for a given IP/WDM network, where $V=(R, O)$, R is the set of integrated router/OXC nodes, O is the set of OXC nodes; L is the set of bidirectional optical links and W is the number of wavelengths per fiber link. The enhanced BIG network model can be obtained from a given network topology G as follows. First, we replicate the original topology of G $|W|$ times. Each copy represents a wavelength and has the same topology as the original IP over WDM network. Then we

check each integrated node $r \in R$. In the enhanced BIG model, node r has $|W|$ copies as $(r_1, r_2 \dots r_{|W|})$. Virtual links $r_1 r_2, r_2 r_3 \dots r_{|W|-1} r_{|W|}$ are then added to connect the corresponding nodes. Notice since those nodes like $(r_1, r_2 \dots r_{|W|})$ are actually the same node, the virtual links only characterize the link connectivity with unlimited bandwidth and have no weight (distance) in the routing and grooming computation. Figure 5(b) is an enhanced BIG network model of the network topology in figure 5(a).

BIH Construction

After the BIG modeling, we need to build more levels of BIH based on different traffic bandwidth requests to facilitate future resource allocation. The most primitive idea is to build a new level once there is a different traffic bandwidth request. Although this method can accurately abstract network resource availability and minimize the search space, the disadvantages are obvious: It may not be responding fast enough to handle large amounts of dynamic traffic. Also, too many levels make the algorithm not scale well. Our idea is to pick up a representative bandwidth according to incoming traffic statistics for a certain period. And the number of levels is also related to the network topology. Given a specific network topology and recent traffic statistics, we predefine the BIH. For example the bandwidth used to build BIG in figure 7 can be predefined as 0.2, 0.5 and 0.8.

Route Existence Check and BI Assigning Procedure

After predefining the proper BIH, when a new traffic request arrives, we pick up the closest BIG level in the BIH to apply routing heuristics. Consider a request $Du = (X_\mu, Y_\mu, \beta_\mu)$ where X_μ and Y_μ are source node and destination node, β_μ is the required bandwidth, using the BI Routing Existence property we immediately know whether the request can be satisfied or not based on a β_μ -BIG without any computing. As we stated before, the given BIH building is not desirable because of time and high maintenance cost. With the predefined limited levels of BIH, it is possible we don't have an exact match of BIG but we can still check the route existence of most requests much faster than a full network search. The route existence screening process is illustrated by an example. Assume a predefined H level BIH $(\alpha_1, \alpha_2 \dots \alpha_H)$, where α_i is the bandwidth level of the corresponding BIG level and $\alpha_1 < \alpha_2 < \dots < \alpha_H$. If β_μ is equal to any predefined bandwidth value α_i the result can be obtained immediately.

If $\beta_\mu > \alpha_H$, we assign Du to α_H -BIG. Then we check whether X_μ and Y_μ are in the same BI of α_H -BIG. If the answer is no, the request is blocked. If yes, we have to do a further check on this BI using Dijkstra's algorithm or a link-state routing protocol.

If $\beta_\mu < \alpha_1$, we assign Du to α_1 -BIG. Then we check whether X_μ and Y_μ are in the same BI of α_1 -BIG. If the answer is yes, the route exists. If no, we have to do a further check on the whole network topology using Dijkstra's algorithm or a link-state routing protocol. This is the worst case in our screening process.

If $\alpha_1 < \beta_\mu < \alpha_H$, say $\alpha_i < \beta_\mu < \alpha_{i+1}$ ($1 \leq i \leq H-1$), we first check whether X_μ and Y_μ are in the same BI of α_{i+1} -BIG. If the answer is yes, the route exists. If not, we then check whether X_μ and Y_μ are in the same BI of α_i -BIG. If they are in the same BI, we have to do a further check on this BI using Dijkstra's algorithm or a link-state routing protocol. If not, the request is blocked.

Consider all the scenarios, except in the worst case we have to check the whole network topology, normally, we can tell the route existence immediately or only need to do searching in a much smaller space. By analyzing the traffic statistics and carefully distributing the BI hierarchy, we can reduce the computation cost significantly and identify the bottleneck links more efficiently.

Grooming

In the IP/WDM integrated routing, it is better to reuse the active lightpaths than create a new lightpath. Our algorithm dynamically combines different traffic flows to efficiently use the network resources by adding a cost function which contemplates the possibility of reusing an already established lightpath to reach the destination. The cost function is represented by the expression:

$$\text{Path_Cost} = N \text{ links} + P * \text{Active_Lightpaths}$$

Where N is the number of new optical links along the path; active lightpaths mean the cut through arcs that have already been set up. P is the cost coefficient that can be adjusted to the best network performance. Usually, P is less than one to reduce the total cost of the path and encourage the usage of active lightpaths. The experimental results in [2] show that based on the cost function, dynamic grooming is achieved automatically assuming that the link state database is properly updated. In our simulation, we find even when P is equal to 1, the path composed of active lightpaths is more likely to be picked if shortest path heuristic is employed. The reason is that the cut through arcs which usually bypass several optical links tend to be shorter.

Integrated Routing and Grooming Algorithm

In the following, we introduce the steps of the integrated routing and grooming algorithm. The traffic requests arrive dynamically and new lightpaths are set up or released as needed. Firstly the network topology is transformed to the enhanced BIG network model. The bandwidth values of different BIH levels are decided based on previous traffic statistics. The incoming traffic request is assigned to the closest hierarchy level according to the bandwidth requirement. We then compute the path in a blocking island holding both the source node and the destination node. The path satisfying the requirement will be shortest in terms of cost function and causes the least impact on the integrity of the whole network. We then compute K different weighted shortest routes. Each route is assigned to different levels of the BIH to compute the splitting number. The splitting happens in the level with the lower bandwidth value does more damage to the integrity of the network than in the level with the high bandwidth value. We here use a cost function to measure the splitting severity. The cost function is represented by the expression:

$$\text{Splitting_Cost} = N_i * \text{the number of splitting in level } i$$

Where N_i is the cost coefficient of hierarchy level i . In BIH, the lower the level i in the hierarchy, the smaller the bandwidth value of BIG is and the more damage will be caused if a splitting happens. So we assign the lower level with higher cost. Assuming there are H levels, we have $N_1 = H, N_2 = H-1 \dots N_H = 1$. The route with minimum splitting cost will be chosen. If several routes have the same splitting cost, we use other heuristics (such as the most loaded link heuristic) or simply select the route randomly.

Once the path is decided, the enhanced BIG network model is modified with the addition of the lightpaths and BIH is updated with the new bandwidth availability on affected links. Notice the above description is about the request arrival. Once a request leaves, the whole procedure needs to be reversed. The bandwidth of affected links is increased and the BIH is updated as needed. If the available bandwidth of an IP logical link (a lightpath in the optical layer) is 1, the logical link is released and the optical links composed of the corresponding lightpath will be reintroduced to the BIG network model.

Below is a formal description of the algorithm: before processing the incoming dynamic traffic, we assume the network has been transformed into an enhanced BIG network with the BIH constructed.

Algorithm: Integrated Routing and Grooming based on BI paradigm

Input: Dynamic traffic request D with bandwidth requirement β ; a BIG network and corresponding BIH

Output: A route satisfying D ; if there is no such a route, D is blocked

Description:

1. Assign D to appropriate BIG of the BIH according to β ;
2. Route existence check; if the route is not available, D is blocked;
3. Compute K different weighted shortest routes using the path cost function;
4. Assigned each of K routes to all levels of BIH to compute the splitting cost;
5. Order K routes and pick up the route(s) with minimum splitting cost;
6. if there are routes with the same minimum splitting cost, choose one randomly or using other heuristics such as the most loaded link heuristic;
7. Update BIG network and the affected blocking islands in BIH;

Complexity analysis

Define a network topology $G(V, L, W)$ for a given IP over WDM network, where V is the set of nodes, L is the set of links and W is the set of wavelengths per fiber link. Assume the set of wavelengths on each fiber link is the same. After transforming into the enhanced BIG network, the number of nodes in the BIG network is $|VW|$ and the predefined BIH level is H . Assume $V=(R, O)$, R is the set of integrated router/OXC nodes, O is the set of OXC nodes. The number of links is equal to $|LW|+|R(|W|-1)|$, $|R(|W|-1)|$ is the number of added virtual links which are regarded as having unlimited bandwidth and no weight. The most common operation in the integrated algorithm is the Blocking Island construction. The β -BI for a given node x of a network can be obtained with a simple greedy algorithm. Starting with an initial set $\{x\}$, we recursively add every node to the set, if this node can be reached from any node in the set by a link that has at least β available bandwidth. In the worst case, this construction process will examine all links. Therefore, the β -BI construction process is linear in $O(n)$, where n is the number of actual links in the network ($n=|LW|$). The time of constructing one level of BIG is $O(mn)$, where m is the number of nodes and n is the number of links in the network ($m=|VW|$, $n=|LW|$). The building time of BIH is $O(Hmn)$. Given a good distribution of BIH levels, usually the route existence check time is equal to the time of constructing one level of BIG. In the worst case, it has to compute the route in the whole enhanced BIG network using Dijkstra's algorithm to decide whether the route is available or not. If the request can be satisfied, the running time is equal to the combination of 1) K alternate shortest paths; 2) Splitting cost computation 3) BIH update. That is $K*O(nlg(m))+K*O(Hmn)+O(Hmn)$, where K is a constant and H is a constant, so the algorithm running time is linear in $O(mn)$.

6. Performance studies

In this section, we will compare the performance of our routing algorithm with the integrated Min-hop routing algorithm (IMH), the Maximum Open Capacity algorithm (MOCA) [2] and the IP-WDM grooming algorithm (IWG) [3]. The main performance measure we use is the blocking probability, which is the ratio of the number of rejected requests to the number of total arriving requests. We employ a random dynamic traffic model to generate the incoming traffic. Calls (requests) arrive at each node according to an independent Poisson process with an arrival rate α . An arriving session is equally likely to be delivered to any node in the network. The session holding time is exponentially distributed with mean $1/\mu$. The bandwidth requirement is uniformly distributed between 0.1 and 1 unit. In our simulation, extensive tests are carried out to ensure a steady state is reached. Because our algorithm assumes the bandwidth requirement of requests can be a fraction of a wavelength, we need to modify the incoming traffic model to limit the source and destination nodes within the set of integrated router/OXC nodes.

As to other heuristics used in the simulation, the integrated Min-hop routing employs the model proposed in [2] to represent IP over WDM networks and the shortest path heuristic is used to compute the route; MOCA computes the critical weight of all links before applying the weighted shortest path heuristic. Notice the MOCA algorithm must have priori knowledge on the ingress/egress nodes; the IP-WDM grooming algorithm aggregates traffic flows on active lightpaths by using a path cost function similar to the one we use. In the simulation, we assume the cost coefficient P is 0.8.

We tested the performances of the algorithm on two different network topologies. The first network is shown in figure 7, which is a NSFNet with 14 nodes and 21 links. The integrated router/OXC nodes are shaded and the nodes that are not shaded are OXCs with no wavelength conversion. The number of wavelengths per link is assumed to be 3. We predefine the BIH with 0.1-BIG, 0.3-BIG, 0.5-BIG and 0.8-BIG. In figure 7, 5 nodes are shaded. Since any two of those 5 nodes can be ingress/egress node pairs, the number of ingress/egress pairs is 10.

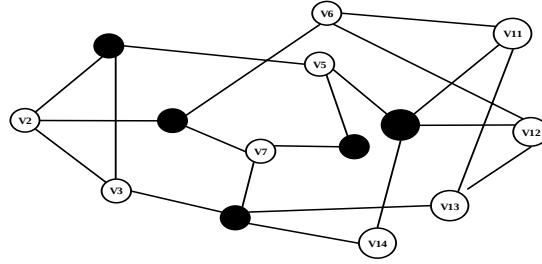


Figure 7: 14 nodes NSFNet with 5 wavelength converters

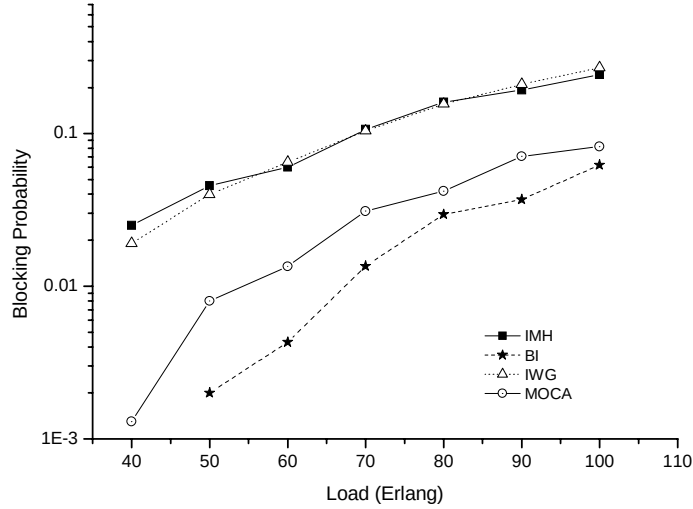


Figure 8: Comparison results of blocking probability for NSFNet

Figure 8 shows the simulation results on the network shown in figure 7. Clearly our integrated algorithm performs much better than the other algorithms. MOCA trails behind. IMH and IWG give the worst performance. For example, when the total load is 90 Erlang, the blocking probability for BI is only 3.7×10^{-2} , compared with 7.1×10^{-2} for MOCA, 1.9×10^{-1} for IMH and 2.1×10^{-1} for IWG. The performance of IWG fluctuates around IMH with the system load increasing. The obvious reason is this algorithm focuses mainly on grooming and sometimes may occupy more resources to achieve the grooming.

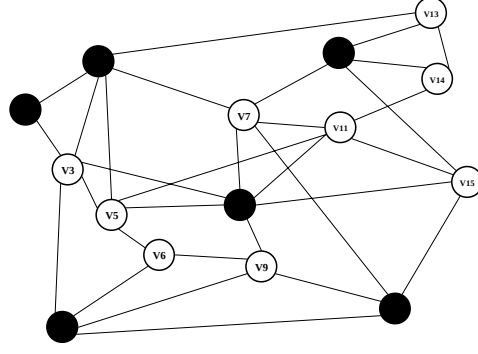


Figure 9: A random generated network topology

The second network topology is shown in figure 9, which is a randomly generated network topology with 15 nodes and 29 links. 6 nodes are chosen as integrated router/OXC nodes so we have 30 pairs of ingress/egress nodes. The number of wavelength is assumed to be 6. We predefine the BIH with 0.1-BIG, 0.3-BIG, 0.5-BIG and 0.8-BIG. The simulation results are shown in figure 10. As expected, the network load capacity increases with the same blocking probability performance for all 4 algorithms. The performance of all algorithms has the similar pattern as in figure 8. Our algorithm outperforms the other algorithms and has a much lower blocking probability, followed by MOCA. The performance of IMH and IWG is similar but lag far behind the previous two algorithms. From the figure, we can see when the network load is light, IWG performs better than IMH, while when the network load is heavy, IMH performs better than IWG.

We also tested our algorithm in a large network with 100 nodes and 130 links. There are 30 integrated router/OXC nodes. The computation time for MOCA is very long since we have got 870 ingress/egress nodes to calculate. It is not practical for MOCA to handle dynamic traffic in this case. In the rest of 3 algorithms, our algorithm still gives the best performance.

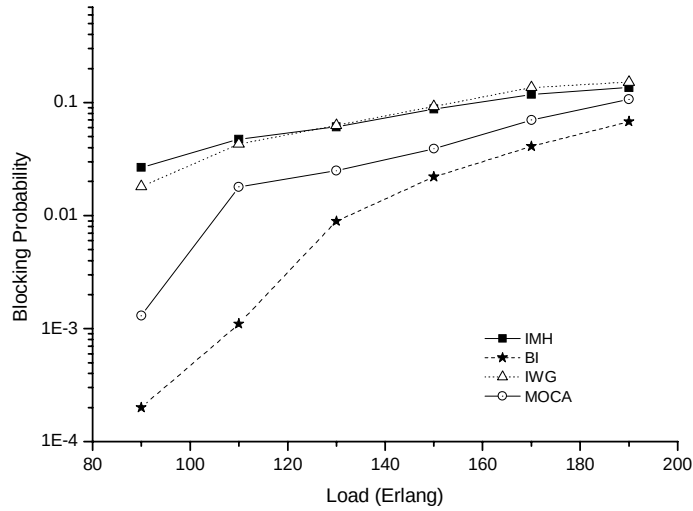


Figure 10: Comparison results of blocking probability for the random network

7. Conclusion

The contribution of this paper is to propose a novel integrated routing and grooming algorithm for routing bandwidth guaranteed paths. To the best of our knowledge, it is the first algorithm that considers routing and grooming together in IP over WDM networks. In this paper, we assume a GMPLS-based peer model to implement the algorithm. Borrowing ideas from a clustering technique called Blocking Island

Paradigm, we propose an enhanced BIG network model to abstract network resources. We discussed in detail the BIH building, route existence check, BI assignment and grooming cost function. A routing heuristic called Minimum Splitting is then developed. The idea is to keep the integrity of blocking islands as intact as possible. The comparison results in the simulation clearly demonstrate that our algorithm has a very good blocking probability performance. We are currently exploring the feasibility of the distributed implementation of the proposed algorithm to even reduce its time complexity further. We are also working on an extension of this algorithm to address the provision/protection problem in IP over WDM networks.

REFERENCE

- [1] B. Rajagopalan *et al*, "IP over Optical Networks: Architectural Aspects," *IEEE Commun. Mag.*, vol. 38 no. 9, Sept. 2000, pp. 94-102.
- [2] J. Comellas *et al*, "Integrated IP/WDM Routing in GMPLS-Based Optical Networks," *IEEE Network*, March/April 2003, pp. 22-27.
- [3] M. Kodialam and T. V. Lakshman, "Integrated Dynamic IP and Wavelength Routing in IP over WDM Networks," *IEEE INFOCOM2001*, pp.358-366.
- [4] C. R. Frei, "Abstraction Techniques for Resource Allocation in Communication Networks," *PhD thesis*, Swiss Federal Institute of Technology – Lausanne, 2000
- [5] D. Zhemina and M. Hamdi, "A Simple and Intelligent Routing and Wavelength Assignment Algorithm for All-Optical Networks," *SPIE Opticom*, 2001, pp. 210-226.
- [6] L. Berger, Ed., "Generalized MPLS – Signaling Functional Description, " Internet draft, draft-ietf-mpls-generalized-signaling-09.txt, 2002, work in progress.
- [7] J. Moy, "OSPF: Anatomy of an Internet Routing Protocol," *Addison-Wesley*, 1998
- [8] A. Banerjee *et al*., "Generalized Multiprotocol Label Switching: An Overview of Signaling Enhancements and Recovery Techniques," *IEEE Commun. Mag.*, vol. 39, no. 7, July 2001, pp. 144-51